

Resolved–Null Covariance Coupling in CTMT Resolution Geometry:

A Stabilizer Invariant, Decomposition Criterion, Protocol-Invariance Battery,
& Conservative Non-Claims

Matěj Rada

Email: MatejRada@email.cz

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Abstract

Chronotopic Metric Theory (CTMT) represents a local observation chart by a resolution geometry

$$\mathcal{G} = (O; R, N, \Sigma),$$

where O is a finite-dimensional observation space, $R = \text{Im } J$ is the resolved observable-response sector, $N = \ker J^\top$ is the unresolved or null sector, and Σ is the covariance structure declared or estimated by the observation protocol. Previous CTMT work classified local admissible morphisms as automorphisms of $\mathcal{G}$. The present paper isolates the off-diagonal covariance block

$$C_{RN} := \Sigma|_{R \times N}$$

as a first-class geometric object.

The main mathematical point is simple but decisive. Relative to $O = R \oplus N$, covariance decomposes as

$$\Sigma = \begin{pmatrix} \Sigma_R & C_{RN} \\ C_{RN}^\top & \Sigma_N \end{pmatrix}.$$

The condition $C_{RN} = 0$ is equivalent to covariance-orthogonality of resolved and null sectors. When $C_{RN} = 0$, the strict automorphism group factors as $O(\Sigma_R) \times O(\Sigma_N)$. When $C_{RN} \neq 0$, the automorphism group shrinks to the stabilizer of C_{RN} inside that product. Thus resolved–null covariance coupling is not a decorative noise term: it changes the morphism class of the local resolution geometry.

Because covariance is protocol-dependent, this paper does not claim that every nonzero C_{RN} is physically intrinsic. Instead, it proposes a protocol-invariance criterion: a nonzero coupling becomes evidence for intrinsic observable geometry only when its orientation survives declared observation morphisms and controlled degradation, while failing under undeclared or inadmissible protocol distortions. A controlled synthetic battery supports this interpretation: declared sensor-basis transformations and gain-rotation transforms preserved C_{RN} to numerical precision; Gaussian degradation perturbed it continuously; undeclared sensor rotation destroyed coupling orientation. The paper closes with falsifiers, limitations, non-claims, and an appendix summarizing the battery.

Contents

1 Purpose and Claim Stratification	2
2 Local Resolution Geometry	3
3 Covariance Block Decomposition	3

4	Vanishing Coupling as a Decomposition Criterion	4
5	Automorphism Stabilizer Theorem	5
6	Relation to Fisher Geometry	6
7	Protocol-Invariance Criterion	6
8	Synthetic Protocol-Invariance Battery	6
8.1	Design	7
8.2	Summary of observed results	7
9	Falsification Criteria	8
10	Non-Claims and Limitations	8
11	Conclusion	9
A	Appendix A: Battery Summary	9
A.1	Reference chart	9
A.2	Compared protocols	9
A.3	Raw diagnostic table	10
A.4	Interpretation rules	10
A.5	Reproducibility note	10
B	Appendix B: Recommended Next Experiments	10

1 Purpose and Claim Stratification

This paper has one narrow purpose: to formalize the role of resolved–null covariance coupling in CTMT resolution geometry.

It does not propose a new covariance estimator. It does not claim that covariance estimation is unique. It does not claim that every observed cross-covariance has physical origin. The purpose is instead to prove that, once a local resolution geometry

$$\mathcal{G} = (O; R, N, \Sigma)$$

has been declared, the block

$$C_{RN} = \Sigma|_{R \times N}$$

is part of the local geometric identity and changes the automorphism group.

Mathematical layer. The paper proves that $C_{RN} = 0$ is exactly covariance-orthogonality of R and N , and that $C_{RN} \neq 0$ changes the automorphism group from a product group to a stabilizer group.

Operational layer. The paper gives computable diagnostics for coupling size, coupling orientation, and protocol stability.

Experimental layer. A controlled synthetic battery tests whether C_{RN} is preserved under declared observation morphisms and degraded under inadmissible protocol changes.

Non-claim layer. The paper does not infer universal physical meaning from nonzero coupling. It only establishes a geometric role and a falsifiable protocol-invariance criterion.

2 Local Resolution Geometry

Let O be a finite-dimensional real observation space. A local observation chart consists of a forward response map

$$F : \Theta \rightarrow O$$

from admissible perturbation variables Θ into observables. At a base point θ_0 , the observable-response Jacobian is

$$J = D_\theta F(\theta_0) : T_{\theta_0} \Theta \rightarrow O.$$

Definition 1 (Resolved sector). The resolved sector is

$$R := \text{Im } J \subset O.$$

Definition 2 (Null sector). Relative to the declared observation pairing, the unresolved or null sector is

$$N := \ker J^\top \subset O.$$

When the pairing is Euclidean, $N = R^\perp$. More generally, N is the annihilator of first-order observable response under the declared pairing.

Definition 3 (Strict resolution geometry). A strict CTMT resolution geometry is

$$\mathcal{G} = (O; R, N, \Sigma),$$

where R and N are the resolved and null sectors, and Σ is the calibrated covariance structure of the observation protocol.

Definition 4 (Conformal resolution geometry). A conformal CTMT resolution geometry is

$$\mathcal{G}_c = (O; R, N, [\Sigma]),$$

where

$$[\Sigma] = \{\lambda \Sigma : \lambda > 0\}.$$

This version is admissible only when the protocol explicitly discards absolute covariance scale.

The local CTMT morphism axiom is that admissible morphisms are automorphisms of the declared resolution geometry:

$$\phi \in \text{Mor}_{\text{CTMT}} \iff \phi^* \mathcal{G} = \mathcal{G}.$$

In matrix form, the strict condition is

$$\phi(R) = R, \quad \phi(N) = N, \quad \phi \Sigma \phi^\top = \Sigma.$$

The conformal condition replaces the last equation by

$$\phi \Sigma \phi^\top = \lambda \Sigma, \quad \lambda > 0.$$

3 Covariance Block Decomposition

Assume an adapted decomposition

$$O = R \oplus N.$$

Let $Q = [Q_R, Q_N]$ be a basis adapted to this decomposition. Then

$$Q^\top \Sigma Q = \begin{pmatrix} \Sigma_R & C_{RN} \\ C_{RN}^\top & \Sigma_N \end{pmatrix}.$$

Definition 5 (Resolved covariance). The resolved covariance block is

$$\Sigma_R := Q_R^\top \Sigma Q_R.$$

Definition 6 (Null covariance). The null covariance block is

$$\Sigma_N := Q_N^\top \Sigma Q_N.$$

Definition 7 (Resolved–null covariance coupling). The resolved–null covariance coupling is

$$C_{RN} := Q_R^\top \Sigma Q_N.$$

Equivalently, C_{RN} is the restriction of the covariance bilinear form to $R \times N$.

Remark 1 (Coordinate representation). The matrix entries of C_{RN} depend on the adapted bases. The statement that the bilinear restriction $\Sigma|_{R \times N}$ vanishes, has a given rank, or is preserved by transport is coordinate-independent.

4 Vanishing Coupling as a Decomposition Criterion

The safest interpretation of nonzero coupling begins with the exact interpretation of zero coupling.

Proposition 8 (Covariance-orthogonality criterion). Let $O = R \oplus N$ and let Σ be a covariance form on O . Then

$$C_{RN} = 0$$

if and only if

$$\Sigma(r, n) = 0 \quad \forall r \in R, n \in N.$$

Proof 9. In an adapted basis $Q = [Q_R, Q_N]$, the cross block is

$$C_{RN} = Q_R^\top \Sigma Q_N.$$

Thus $C_{RN} = 0$ precisely when every covariance pairing between a resolved vector and a null vector vanishes. This is exactly Σ -orthogonality of R and N .

Corollary 10 (Direct-sum covariance geometry). If $C_{RN} = 0$, then

$$\Sigma = \Sigma_R \oplus \Sigma_N$$

relative to $O = R \oplus N$. The resolved and null covariance geometries then form independent second-order sectors.

Corollary 11 (Nonzero coupling as failure of covariance decomposition). If $C_{RN} \neq 0$, then the resolved and null sectors do not form a covariance-orthogonal decomposition. Nonzero coupling measures the failure of the resolved/null split to be a direct covariance decomposition.

Remark 2 (What this does and does not say). The statement $C_{RN} \neq 0$ does not by itself identify a physical cause. It says that, relative to the declared observation protocol, resolved and unresolved directions are second-order coupled. Whether that coupling arises from physical transport, instrumentation, sampling, preprocessing, or estimator bias is an experimental and protocol question.

5 Automorphism Stabilizer Theorem

The geometric importance of C_{RN} is that it enters the automorphism group.

Lemma 12 (Sector preservation gives block form). If $\phi(R) = R$ and $\phi(N) = N$, then in an adapted decomposition

$$\phi = \text{diag}(A, B),$$

where $A : R \rightarrow R$ and $B : N \rightarrow N$.

Proof 13. Sector preservation forbids components from R into N and from N into R . Hence ϕ is block diagonal in the adapted decomposition.

Theorem 1 (Resolved-null coupling stabilizer). Let

$$\mathcal{G} = (O; R, N, \Sigma)$$

be a strict CTMT resolution geometry with

$$\Sigma = \begin{pmatrix} \Sigma_R & C_{RN} \\ C_{RN}^\top & \Sigma_N \end{pmatrix}.$$

Then

$$\text{Aut}(\mathcal{G}) = \left\{ \text{diag}(A, B) : A\Sigma_R A^\top = \Sigma_R, B\Sigma_N B^\top = \Sigma_N, AC_{RN}B^\top = C_{RN} \right\}.$$

If $C_{RN} = 0$, then

$$\text{Aut}(\mathcal{G}) \cong O(\Sigma_R) \times O(\Sigma_N).$$

If $C_{RN} \neq 0$, then $\text{Aut}(\mathcal{G})$ is the stabilizer of C_{RN} inside

$$O(\Sigma_R) \times O(\Sigma_N).$$

Proof 14. By the preceding lemma, an automorphism has block form $\phi = \text{diag}(A, B)$. Substitution into $\phi\Sigma\phi^\top = \Sigma$ gives

$$\begin{pmatrix} A\Sigma_R A^\top & AC_{RN}B^\top \\ BC_{RN}^\top A^\top & B\Sigma_N B^\top \end{pmatrix} = \begin{pmatrix} \Sigma_R & C_{RN} \\ C_{RN}^\top & \Sigma_N \end{pmatrix}.$$

Equating blocks gives the three defining equations. If $C_{RN} = 0$, the cross-block equation is vacuous and the group factors. If $C_{RN} \neq 0$, only pairs (A, B) stabilizing C_{RN} remain.

Corollary 15 (Coupling changes morphism class). Two charts may have the same R , N , Σ_R , and Σ_N , but different C_{RN} . Such charts can have different automorphism groups. Therefore C_{RN} is a morphism-class invariant of strict CTMT resolution geometry.

Corollary 16 (Conformal coupling stabilizer). For conformal CTMT geometry

$$\mathcal{G}_c = (O; R, N, [\Sigma]),$$

the automorphism group is

$$\text{Aut}(\mathcal{G}_c) = \left\{ \text{diag}(A, B) : \exists \lambda > 0, A\Sigma_R A^\top = \lambda\Sigma_R, B\Sigma_N B^\top = \lambda\Sigma_N, AC_{RN}B^\top = \lambda C_{RN} \right\}.$$

Sector-only rescalings are not conformal CTMT symmetries unless a separate sector-relative protocol is explicitly declared.

6 Relation to Fisher Geometry

Given positive definite Σ , the Fisher pullback is

$$G_F = J^\top \Sigma^{-1} J.$$

This object is central in statistical and information geometry [3–5]. CTMT does not replace Fisher geometry. Instead, it treats Fisher geometry as a derived diagnostic of the larger resolution geometry.

Proposition 17 (Fisher insufficiency). The Fisher pullback G_F does not determine the full covariance geometry

$$(R, N, \Sigma_R, \Sigma_N, C_{RN}).$$

Proof 18. The Fisher pullback is defined on perturbation space through the resolved response directions $Jv \in R$. Covariance changes supported in N , or changes in resolved–null coupling that do not alter the pullback, need not be visible in G_F . Hence G_F is not sufficient to reconstruct the full covariance geometry on O .

Remark 3 (Why coupling cannot be dropped). If C_{RN} is dropped, then distinct resolution geometries may be identified even though their automorphism groups differ. A Fisher-only or block-diagonal-only invariant can therefore produce false-positive similarity.

7 Protocol-Invariance Criterion

Because covariance is measurement-dependent, the physical interpretation of C_{RN} cannot be obtained from nonzero magnitude alone. The correct question is whether the coupling survives independent declared observation protocols.

Principle 2 (Protocol-invariance criterion). A resolved–null coupling C_{RN} is evidence for intrinsic observable-geometric coupling only if it remains stable under declared observation morphisms and controlled degradation, while changing under undeclared or inadmissible protocol distortions.

Declared observation morphism. If an observation coordinate transformation $T : O \rightarrow O$ is declared, then the transported chart uses

$$J \mapsto TJ, \quad \Sigma \mapsto T\Sigma T^\top.$$

The coupling must be compared after this transport is accounted for.

Undeclared protocol distortion. If observations are transformed but the protocol does not declare the transport, then the comparison is not between equivalent charts. Such a distortion may create or destroy apparent coupling.

Controlled degradation. Noise, missingness, temporal jitter, and estimator variation should perturb C_{RN} continuously when the geometry is sufficiently reconstructable. Abrupt or unstable changes indicate either insufficient data or protocol corruption.

8 Synthetic Protocol-Invariance Battery

A controlled synthetic battery was performed to test protocol behavior of a known nonzero C_{RN} . The aim was not to prove physical universality. The aim was to verify whether C_{RN} behaves like a resolution-geometric invariant under declared transports, and whether inadmissible protocol changes are detected.

8.1 Design

A reference chart was generated with fixed J , fixed $R = \text{Im } J$, fixed $N = \ker J^\top$, and controlled nonzero coupling C_{RN} . Synthetic sample clouds were drawn from the corresponding covariance geometry. The same underlying geometry was then evaluated under:

- (a) a declared orthogonal sensor-basis transformation;
- (b) a declared gain-and-rotation transformation;
- (c) low Gaussian degradation;
- (d) high Gaussian degradation;
- (e) undeclared sensor rotation;
- (f) artificial cross-bias contamination.

For each protocol, the estimated coupling was compared with the reference coupling by

$$d_C = \frac{\|C_{RN}^{(0)} - C_{RN}^{(1)}\|_F}{\|C_{RN}^{(0)}\|_F + \|C_{RN}^{(1)}\|_F},$$

and by the Frobenius cosine

$$\cos(C_{RN}^{(0)}, C_{RN}^{(1)}) = \frac{\langle C_{RN}^{(0)}, C_{RN}^{(1)} \rangle_F}{\|C_{RN}^{(0)}\|_F \|C_{RN}^{(1)}\|_F}.$$

8.2 Summary of observed results

Protocol	d_C vs. base	coupling cosine	Verdict
Declared orthogonal sensor basis	1.95×10^{-15}	1.000000	stable
Declared gain and rotation	3.04×10^{-15}	1.000000	stable
Low Gaussian degradation	8.53×10^{-3}	0.999854	stable
High Gaussian degradation	3.12×10^{-2}	0.998056	stable
Undeclared sensor rotation	7.50×10^{-1}	0.006344	rejected
Estimator cross-bias artifact	5.39×10^{-2}	0.994191	ambiguous

Table 1: Synthetic protocol-invariance battery for resolved-null covariance coupling. Declared transports preserved coupling to numerical precision. Gaussian degradation changed the estimate continuously. Undeclared sensor rotation destroyed coupling orientation. The artificial cross-bias case remained ambiguous, illustrating that apparent coupling stability alone is insufficient to infer physical origin.

Proposition 19 (Synthetic protocol-invariance observation). In the controlled synthetic battery, C_{RN} was preserved to numerical precision under declared observation transports, changed continuously under Gaussian degradation, and failed under undeclared sensor rotation.

Proof 20 (Numerical verification). Declared orthogonal sensor-basis transport gave

$$d_C = 1.95 \times 10^{-15}, \quad \cos(C_{RN}^{(0)}, C_{RN}^{(1)}) = 1.000000.$$

Declared gain-and-rotation transport gave

$$d_C = 3.04 \times 10^{-15}, \quad \cos(C_{RN}^{(0)}, C_{RN}^{(1)}) = 1.000000.$$

Low and high Gaussian degradation gave

$$d_C = 8.53 \times 10^{-3}$$

and

$$d_C = 3.12 \times 10^{-2},$$

while maintaining coupling cosines above 0.998. Undeclared sensor rotation gave

$$d_C = 7.50 \times 10^{-1}, \quad \cos(C_{RN}^{(0)}, C_{RN}^{(1)}) = 0.006344.$$

Therefore coupling behaved as a declared-protocol invariant in the controlled setting and failed under an inadmissible protocol change.

Remark 4 (Meaning of the ambiguous artifact case). The artificial cross-bias case produced relatively small coupling distance and high coupling cosine. This is not a success claim. It is an important warning: some artifacts can mimic stable coupling. Therefore nonzero or apparently stable C_{RN} must be tested across independent protocols before receiving physical interpretation.

9 Falsification Criteria

Falsifier 3 (Protocol instability). If independent declared observation protocols measuring the same transport object produce incompatible coupling orientation, then C_{RN} should not be interpreted as intrinsic observable geometry.

Falsifier 4 (Estimator dependence). If C_{RN} appears only for one estimator, preprocessing method, sample window, filtering pipeline, or covariance regularizer, then it is an estimator artifact until shown otherwise.

Falsifier 5 (Declared-coordinate failure). If a coupling changes orientation under a declared coordinate transformation after pulling the chart back correctly, then the coupling is not a CTMT invariant.

Falsifier 6 (Finite-sample instability). If repeated windows produce unstable C_{RN} while R , N , and covariance scale remain otherwise stable, then coupling is not reliably reconstructable.

Falsifier 7 (False physical reading). If $C_{RN} \neq 0$ is interpreted as physical without independent protocol recovery, the interpretation exceeds the evidence.

10 Non-Claims and Limitations

Non-claim 21 (No automatic physical interpretation). This paper does not claim that every nonzero C_{RN} is physically real. A nonzero cross block may be produced by physics, instrumentation, preprocessing, sampling, estimator bias, or undeclared coordinates.

Non-claim 22 (No new covariance estimator). The paper does not replace established covariance estimation. It studies the geometric role of covariance after a covariance model has been declared or estimated.

Non-claim 23 (No universal validation). The synthetic battery does not prove that nature realizes CTMT resolution geometry. It demonstrates that the coupling behaves correctly under controlled declared and undeclared protocol variations.

Non-claim 24 (No global theorem). Local coupling invariance does not by itself prove global transport closure, holonomy classification, or bundle-level consistency.

Non-claim 25 (No Fisher contradiction). The result does not contradict Fisher information geometry. Fisher geometry remains a derived local diagnostic. The claim is only that Fisher geometry is not sufficient to identify full CTMT covariance geometry.

11 Conclusion

Resolved–null covariance coupling is mathematically locked into CTMT resolution geometry by two facts.

First,

$$C_{RN} = 0 \iff R \perp_{\Sigma} N.$$

Thus zero coupling is exactly covariance-orthogonal decomposition of resolved and null sectors.

Second,

$$C_{RN} \neq 0 \implies \text{Aut}(\mathcal{G}) \text{ is a coupling stabilizer rather than a direct product.}$$

Thus nonzero coupling changes the morphism class.

The correct conservative interpretation is therefore:

C_{RN} is not automatically a physical invariant, but it is a genuine invariant of the local resolution geometry under admissible observation transport.

The synthetic protocol battery supports this interpretation. Declared transports preserved coupling to numerical precision. Gaussian degradation perturbed coupling continuously. Undeclared sensor transformation destroyed coupling orientation. The remaining frontier is not whether C_{RN} belongs in the local geometry; it does. The remaining question is when repeated independent observation protocols justify interpreting it as a physical transport feature rather than a protocol artifact.

A Appendix A: Battery Summary

This appendix records the controlled protocol-invariance battery used in Section 7.

A.1 Reference chart

A synthetic chart was generated with:

- observation dimension $n = 48$,
- resolved rank $r = 4$,
- null dimension $n - r = 44$,
- controlled nonzero resolved–null covariance coupling,
- sample count $W = 25000$.

The large sample count was used to isolate protocol behavior from finite-sample cross-block noise.

A.2 Compared protocols

The following protocol variations were tested:

- (a) reference protocol;
- (b) declared orthogonal sensor-basis transformation;
- (c) declared gain-and-rotation transformation;

- (d) low Gaussian degradation;
- (e) high Gaussian degradation;
- (f) undeclared sensor rotation;
- (g) artificial estimator cross-bias.

A.3 Raw diagnostic table

Protocol	cross rel.	d_C	cosine	$d\Sigma_R$	$d\Sigma_N$
Base	0.110854	0.000000	1.000000	0.000000	0.000000
Declared orthogonal basis	0.110854	1.95×10^{-15}	1.000000	1.74×10^{-15}	6.69×10^{-16}
Declared gain and rotation	0.110854	3.04×10^{-15}	1.000000	3.52×10^{-15}	7.14×10^{-16}
Low Gaussian degradation	0.109771	0.008534	0.999854	0.003226	0.009949
High Gaussian degradation	0.101032	0.031181	0.998056	0.027862	0.087433
Undeclared sensor rotation	0.236591	0.750150	0.006344	0.577264	0.485541
Estimator cross-bias	0.098956	0.053894	0.994191	0.135091	0.001698

Table 2: Raw diagnostics from the synthetic coupling protocol-invariance battery.

A.4 Interpretation rules

A coupling was classified as stable when:

$$\cos(C_{RN}^{(0)}, C_{RN}^{(1)}) > 0.95$$

and

$$d_C < 0.10$$

under a declared protocol.

A coupling was classified as degraded but surviving when Gaussian degradation preserved orientation while increasing block residuals continuously.

A coupling was classified as artifact or undeclared protocol failure when coupling orientation collapsed or when comparison was made without applying the declared transport.

A.5 Reproducibility note

The battery is synthetic and controlled. It should be treated as a falsification harness, not as empirical validation. Its role is to demonstrate that the diagnostic can distinguish declared transport invariance from undeclared protocol distortion in a known setting.

B Appendix B: Recommended Next Experiments

The next experimental target is independent-protocol recovery. Suitable tests include:

1. the same coil observed with different sensor layouts and bandwidths;
2. the same Navier–Stokes simulation observed through velocity, vorticity, pressure, and energy-density channels;
3. the same seismic region observed through independent catalogues and preprocessing pipelines;

4. deliberate degradation by missing samples, temporal jitter, low-pass filtering, and estimator changes.

The target quantity is not merely

$$\|C_{RN}\|_F.$$

The target is coupling orientation and transport stability across independent admissible protocols.

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